EATDer: Edge-Assisted Adaptive Transformer Detector for Remote Sensing Change Detection

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***Abstract*— Change detection (CD) is one of the important research topics in remote sensing (RS) image processing. Recently, convolutional neural networks (CNNs) have dominated the RSCD community. Many successful CNN-based models have been proposed, and they achieved cracking performance. Nevertheless, influenced by the limited receptive field, the CNN-based models are not good at capturing long-distance context dependencies within RS images, negatively impacting their performance. With the appearance of the visual transformer, the above problems have been mitigated. However, the high time costs of the transformer-based models limit their applicability. In addition, previous CD networks (whether CNN-based or transform-based) do not pay attention to the edges of changed areas, reducing the quality of change maps. To overcome the shortcomings discussed above, we propose a new CD method named edge-assisted adaptive transformer detector (EATDer). EATDer consists of a Siamese encoder and an edge-aware decoder. Each branch in the Siamese encoder encloses three self-adaption vision transformer (SAVT) blocks, which aim to capture the local and global information within RS images. Also, two branches are connected by full-range fusion modules (FRFMs), which focus on mining the temporal clues among bi-temporal RS images and pointing out the changed/unchanged messages. The edge-aware decoder first integrates the multiscale features obtained by the encoder using a restoring block. Then, it enhances the combined features by a refining block. Finally, based on the refined features, both the change and edge detection results can be produced. Along with a joint loss function, we can get high-quality change maps in which the changed areas are correct and have clear and smooth edges. The usefulness of our EATDer is validated by extensive experiments conducted on three popular RSCD datasets. Our source codes are available at https://github.com/TangXu-Group/Remote-Sensing-Image-Change-Detection/tree/main/EATDer**

***Index Terms*— Change detection (CD), remote sensing (RS), transformer.**

I. Introduction

THANKS to the development of sensor technologies, a large number of remote sensing (RS) images can be collected every day. These Earth observation data provide rich land-cover information for researchers. As a basic and vibrant interpretation technique, RS change detection (RSCD) plays a vital role in the RS community. It aims to find the changed pixels and regions within the multitemporal RS images, which cover the same areas but are produced at different times [1], [2]. The accurate RSCD results contribute to many RS applications, including land-cover analysis [3], object detection [4], disaster monitoring [5], and scene understanding [6], [7], [8], [9].

In the past, numerous methods have been developed for RSCD with the help of traditional machine learning [10], [11]. Those methods mainly rely on algebra and transformation techniques. The algebra-based methods, such as change vector analysis [12], are used to determine the changed pixels by performing the common algebraic operations on the original image space. The transformation-based approaches, such as principal component analysis [13], are habituated to mapping RS images into suitable feature spaces for deciding the changed areas. Although the above methods are feasible, their performance cannot meet what we expected due to the hand-crafted features, which are not able to represent the complex contents within RS images. For example, the pseudo variation issues (see Fig. 1) caused by illumination and shadows are hardly addressed using only the low-level visual features.

Recently, with the appearance of deep learning, especially convolutional neural networks (CNNs), the results of RSCD have improved significantly. Benefiting from the stacked structure and hierarchical learning manner, CNNs can explore the semantic features of RS images. Compared with the hand-crafted features, these high-level visual features can fully express intricate land covers within RS images so that accurate RSCD maps can be generated [14]. Many CNN-based models have been proposed, and their performance is cracking [15], [16]. In the beginning, scholars adopt some existing CNNs to construct their RSCD models [17]. Then, some RSCD-oriented CNNs are developed [18] to enhance the performance. With the help of some advanced technologies, such as Siamese architecture [19] and attention mechanisms [20], [21], these RSCD-oriented CNNs promote the RSCD results to a new level. Despite great successes, the CNN-based RSCD methods still have an inherent problem, that is, they are not able to capture the long-range contextual information hidden in RS images. This shortcoming limits their behavior more or less. For instance, a set of bi-temporal RS images is shown in Fig. 2(a) and (b). CNNs can capture local information (e.g., "trees," "cars," and "buildings") easily. If we only leverage the local clues, the generated change map may contain misjudgments (see Fig. 2(c) in the first row, areas in red). If the global relationships between various local objects can be further

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(a) (b)

Fig. 1. Two common interference factors of RSCD task. (a) Caused by illumination. (b) Caused by shadow.



(a) (b) (c)

Fig. 2. Examples of how local and global information influence the RSCD results. (a) Former RS images. (b) Later RS images. (c) RSCD maps. CNNs are good at capturing the local land covers. If we only use the local information, the incorrect RSCD map may be produced. This case is shown in the first row. If the global relations between different local land covers are modeled, the correct RSCD map can be generated. This case is displayed in the second row.

mined, the correct change map can be produced (see Fig. 2(c) in the second row).

Considering the above limitations, researchers introduce the transformer [22], [23] into the RS community. Due to the energetic capacity of long-range knowledge mining, the transformer has become increasingly popular for RS image processing. Various transformer-based models have been developed for different RS tasks [24], [25], as well as RSCD [26], [27]. Although those transformer-based models have shown their superiorities, there is still room for improvement, especially for RSCD tasks. First, due to the multihead self-attention (MSA) mechanism [23], the transformer-based RSCD models' time consumption is always high, which limits their practicality and generalization capability. Second, the complicated content information and temporal cues of different RS images are equally important for RSCD tasks. However, most current transformer-based models focus on the RS images themselves but ignore the temporal clues between RS images collected at different times. Third, since the land covers within RS images are diverse, and their scales are various, the changed areas among multitemporal RS images are usually irregular. However, the normal transformer-based models are not good at exploring the boundaries precisely, which impacts the RSCD tasks negatively.

To overcome the mentioned shortcomings, we develop a new encoder–decoder RSCD model named edge-assisted adaptive transformer detector (EATDer). The encoder is a Siamese network that involves self-adaption vision transformer (SAVT) blocks and full-range fusion modules (FRFMs). The function of SAVT blocks is to explore both the local and global information from RS images with relatively low time consumption. The goals of FRFMs are to mine the invisible temporal knowledge among different RS images and to emphasize the changed/unchanged areas. The decoder contains a restoring block and a refining block. Their targets are to integrate and enhance the multiscale features obtained from encoders and produce the change maps. Meanwhile, an auxiliary edge detection task is added to ensure the decoder is aware of the edges of changed areas so that the change maps' quality can be further enhanced. Our contributions can be summarized as follows.

1) An encoder–decoder RSCD network (i.e., EATDer) is developed. Not only the local details and global context within a single RS image but also the temporal clues hidden in a series of RS images can be explored by EATDer. Therefore, promising CD maps can be generated.
2) Instead of adopting the original vision transformer and its existing variants, we develop an SAVT block to explore the valuable information from RS images. To mine the complex local details within RS images, the common grid patch scheme used in the original transformer is replaced by the overlapping embedding scheme. To reduce computational costs, a self-adaptive MSA (SAMSA) embedded in self-adaption progress (SAP) is designed. In this fashion, local and global messages can be fully captured from RS images with modest time costs.
3) In addition to focusing on the intricate contents of RS images, our EATDer also considers the temporal clues among different RS images to address CD tasks. To this end, we develop a FRFM. Adding it into the encoder allows helpful information across the bi-temporal RS images to be exchanged and fused mutually, and the invisible temporal knowledge among different RS images can be explored. Therefore, the changed/unchanged areas among different RS images can be highlighted.
4) Apart from detecting the changes, we also recognize the edges around the changed areas to enhance the CD results further. For this purpose, the output of the decoder is split at the channel dimension to represent the confidence degrees of the change and edge information. Also, two sets of edge detection ground truths are constructed automatically to assist in identifying the edges.

The rest parts of this article are organized as follows. Section II reviews some literature related to RSCD. Section III explains our EATDer in detail. The experiments and corresponding discussion are displayed in Section IV. Finally, a short conclusion is summarized in Section V.

II. Related Work

A. CNN-Based CD Models

Recently, CNN-based methods have dominated the RSCD community. We roughly divide them into two groups according to their architecture [28], [29], i.e., single- and dual-branch models. In the first group, the methods are used to concatenate the bi-temporal RS images to be the input data for

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the following feature learning and change map generation. Liu et al. [30] proposed a local restricted CNN (LRCNN) to deal with PolSAR image CD tasks. LRCNN receives the combination of the bi-temporal PolSAR images and their difference maps (DIs). Then, by leveraging regression pretraining and classification fine-tuning, the positive change maps can be generated in an unsupervised manner. An end-to-end CNN framework was developed for hyperspectral image (HSI) CD [31]. Considering the high-dimension properties of HSIs, the endmember analysis approach is first adopted to mine cross-channel features and explore multisource information. Then, the preprocessed HSIs are combined and input into a specific CNN for the change maps. Another HSI CD network was presented in [32], in which the self-calibrated convolution is adopted to enhance the features' discrimination and reduce the model's volume simultaneously. Based on the pseudo label generation scheme, fuzzy clustering algorithm, and CNN, a single-branch SAR image CD model was proposed in [33]. The high-confidence samples and their pseudo labels are used to train CNN, and then the trained CNN is used to produce the final change maps.

With the help of booming Siamese networks, many dual-branch models have been proposed for RSCD. In contrast to the single-branch models, the bi-temporal RS images are fed into two branches individually for the following steps. Chen et al. [34] proposed a Siamese convolutional multilayer recurrent neural network (SiamCRNN) for RSCD. By embedding the CNN and recurrent neural network (RNN) into a Siamese framework, both the structural and spatial–spectral information can be explored from multitemporal RS images. Thus, favorable change maps can be obtained. Similarly, another dual-branch network was developed in [35]. The positive CD results are ensured by the extracted local-global features and cross-task transfer learning strategy. Chen et al. [36] designed a dual encoder–encoder network to deal with RSCD tasks, in which the dense connection scheme and self-supervised learning are integrated to generate the encouraging change maps. Apart from constructing various models under the framework of the Siamese networks, some scholars pay attention to capturing the boundary clues for improving RSCD results. Bai et al. [37] presented a dual-branch network. Besides the common Siamese feature learning and difference information extraction, the prior knowledge of the edge structure is also considered to guarantee the accuracy of change maps. Lei et al. [38] added a boundary extraction constraint into the Siamese network to enhance the discrimination of learned visual features and improve the CD results.

B. Transformer-Based CD Models

Transformer, proposed in natural language processing [22], has become popular in the RS community due to its intrinsic capacity in long-range relations exploration [39], [40], [41]. For RSCD, many transformer-based methods have been proposed, and we broadly divide them into two classes.

The methods within the first-class focus on taking advantage of the original transformer to design new models for RSCD. Chen et al. [42] proposed a bi-temporal image transformer (BIT) to handle the RSCD problem by exploring the spatial–temporal contexts. BIT first maps RS images into high-level feature maps using two conventional CNNs. Then, it adopts a transformer encoder to model the contextual relations in both spatial and temporal domains. Finally, BIT leverages two transformer encoders to produce the refined feature maps, which can be used to decide change maps. The encouraging results confirm the effectiveness of the transformer in RSCD. By integrating the superiorities of CNN and transformer, an intrascale cross-interaction and interscale feature fusion network (ICIF-Net) was developed in [43]. ICIF-Net utilizes a CNN and a transformer to capture diverse information from bi-temporal RS images. Along with the intrascale clue interaction and interscale knowledge fusion strategies, the feature maps rich in local-global messages can be obtained to produce RSCD results. Li et al. [44] embedded the transformer into the UNet to develop an encoder–decoder model for RSCD. The UNet is used to capture the detailed local information, while the transformer is used to model the global contextual relations. After combining them, the positive change maps can be predicted. Considering the multiscale properties of RS images, a new model named MTCNet was introduced in [45], in which the conventional CNN, transformer, multiscale information fusion module, and block attention module are merged. The reported CD results demonstrate MTCNet's usefulness to RSCD.

Although the above methods have succeeded, their computational costs are always high because of the self-attention mechanism within the original transformer. To mitigate this problem, the methods within the second class are accustomed to introducing the variants of the transformer for RSCD models. Zhang et al. [27] developed a Siamese U-shaped model to deal with RSCD tasks named SwinSUNet. SwinSUNet consists of two weight-shared encoders, a decoder, and proper feature fusion/merging schemes. The basic unit within the encoder and decoder is the Swin transformer (SwinT) block rather than the common transformer. This way, the effectiveness and efficiency of SwinSUNet can be ensured. Bandara and Patel [46] embedded the sequence reduction into the original transformer so that the time costs of the whole method can be saved. Wang et al. [47] adopted ViTAE [48] to complete RSCD tasks. ViTAE's computational complexity is lower than the original transformer due to the spatial pyramid reduction and shiftable window multihead attention. Thus, the process of RSCD is efficient.

III. Proposed Method

A. Overall Framework

The framework of the proposed EATDer is shown in Fig. 3. It mainly consists of a Siamese encoder and an edge-aware decoder. The Siamese encoder has two same networks, and each of them involves three SAVT blocks to learn valuable information from bi-temporal RS images with a tolerable computational cost. In addition, two networks are connected by our FRFM. In this way, not only can the profitable information corresponding to bi-temporal RS images be transferred, but

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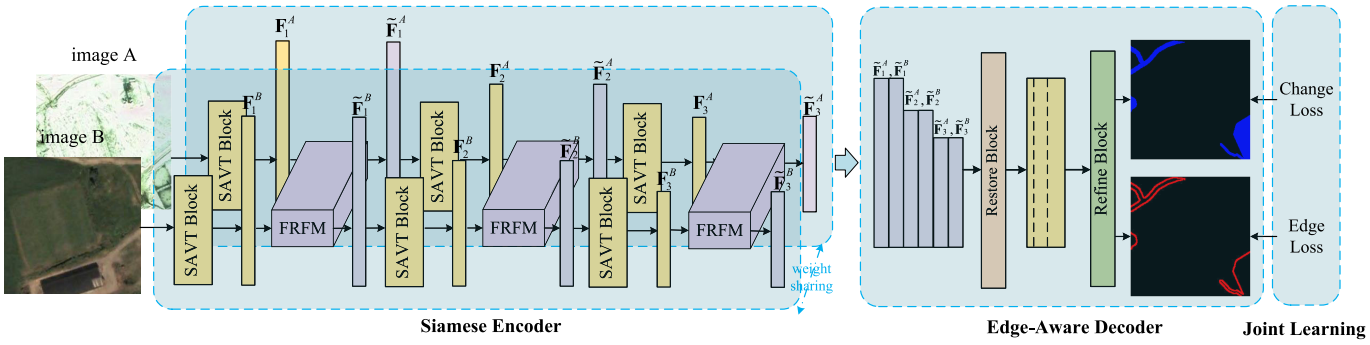


Fig. 3. Framework of the proposed EATDer.

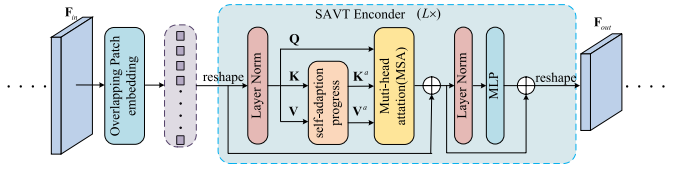


Fig. 4. Architecture of the SAVT block.

also the changed and unchanged clues can be highlighted. The edge-aware decoder mainly incorporates a restoring block and a refining block, which aim to integrate knowledge from various scales and improve the feature maps for CD. In addition, besides generating the change maps, we also estimate the edges of the changed areas. This way, our EATDer could accurately capture the edges of changed land covers to enhance the quality of change maps.

B. Siamese Encoder

This section will explain the Siamese encoder. Before introducing it in detail, we first clarify the SAVT block and FRFM.

1) SAVT Block: The goal of the SAVT block is to capture the complex contents within RS images from a global aspect with an acceptable computational cost. To this end, our SAVT block is developed based on the original vision transformer [23]. Also, we introduce the overlapping embedding scheme [46], [49], [50], [51] to explore the local clues hidden in RS images and remold the MSA mechanism [22], [23] to reduce computation.

The block architecture is shown in Fig. 4. Suppose the input of a SAVT block is $\mathbf{F}_{\text{in}} \in \mathbb{R}^{h\times w\times c}$. Considering the complex and rich local contents within RS images, the overlapping embedding scheme instead of the grid strategy is adopted to transform $\mathbf{F}_{\text{in}}$ into data patches. Particularly, we pass $\mathbf{F}_{\text{in}}$ through a convolution with c' kernels to generate a set of data patches $\mathbf{P} \in \mathbb{R}^{(h/s)\times(w/s)\times c'}$. The sizes, stride, and padding sizes of c' kernels are $2s-1$, s, and $s-1$. Then, the obtained data patches will be fed into L SAVT encoders to generate the updated patches $\mathbf{P}^{\text{new}} \in \mathbb{R}^{(hw/s^2)\times c'}$. Finally, the output $\mathbf{F}_{\text{out}} \in \mathbb{R}^{(h/s)\times(w/s)\times c'}$ of the SAVT block can be obtained after the reshape operation.

Since the L SAVT encoders are the same, we adopt the first SAVT encoder as an example for description. The SAVT encoder involves a SAMSA part and a multilayer perceptron (MLP) part. The layer-norm (LN) layer is added before each part, while the residual connection is adopted after each part. Therefore, the output of the first SAVT encoder is

$$\begin{aligned}\mathbf{P}_1^{\text{SAMSA}} &= \text{SAMSA}\big(\text{LN}\big(\mathbf{P}_1'\big)\big) + \mathbf{P}_1' \\ \mathbf{P}_1^{\text{new}} &= \text{MLP}\big(\text{LN}\big(\mathbf{P}_1^{\text{SAMSA}}\big)\big) + \mathbf{P}_1^{\text{SAMSA}}\end{aligned} \tag{1}$$

where $\mathbf{P}_1'$ denotes the reshaped data patches $\mathbf{P}$ and its size is $(hw/s^2)\times c'$, LN$(\cdot)$, SAMSA$(\cdot)$ and MLP$(\cdot)$ indicate the functions of the LN, SAMSA, and MLP layers, and $\mathbf{P}_1^{\text{new}} \in \mathbb{R}^{(hw/s^2)\times c'}$ means the encoded features. Here, the MLP part contains two linear transformations and a Gaussian error linear unit activation. $\mathbf{P}_1^{\text{new}}$ would be fed into the second SAVT encoder to generate $\mathbf{P}_2^{\text{new}}$. In a similar fashion, we can obtain the updated patches $\mathbf{P}^{\text{new}}$ after L SAVT encoders.

The SAMSA part mainly contains a SAP and a conventional MSA. Here, the "self-adaption" in SAP mainly describes the adaptive spatial size changes of the pending data for MSA. First, we copy the normalized $\mathbf{P}_1'$ to produce pending data $\mathbf{Q}_1 \in \mathbb{R}^{(hw/s^2)\times c'}$, $\mathbf{V}_1 \in \mathbb{R}^{(hw/s^2)\times c'}$, and $\mathbf{K}_1 \in \mathbb{R}^{(hw/s^2)\times c'}$. Customarily, the MSA mechanism can be applied to $\{\mathbf{Q}_1, \mathbf{K}_1, \mathbf{V}_1\}$ directly. However, the computational costs would be large. To mitigate this issue, we develop SAP to reduce the spatial sizes of $\{\mathbf{K}_1, \mathbf{V}_1\}$. For $\mathbf{K}_1$, we reshape it into 2-D version $\tilde{\mathbf{K}}_1 \in \mathbb{R}^{(h/s)\times(w/s)\times c'}$, and use the average pooling and a 1×1 convolution to get $\hat{\mathbf{K}}_1 \in \mathbb{R}^{\lceil\sqrt{(h/s)}\rceil\times\lceil\sqrt{(w/s)}\rceil\times c'}$. Here, the average pooling aims to reduce the spatial size, and the 1×1 convolution avoids the information loss. After the reshape operation, we can get the reductive data $\mathbf{K}_1^a \in \mathbb{R}^{\lceil\sqrt{hw}/s\rceil\times c'}$. Similarly, we can obtain the reductive data of $\mathbf{V}_1$, i.e., $\mathbf{V}_1^a \in \mathbb{R}^{\lceil\sqrt{hw}/s\rceil\times c'}$. Second, the MSA mechanism is conducted on $\{\mathbf{Q}_1, \mathbf{K}_1^a, \mathbf{V}_1^a\}$, which can be formulated as

$$\begin{aligned}\text{MSA}\big(\mathbf{Q}_1, \mathbf{K}_1^a, \mathbf{V}_1^a\big) &= \text{concat}(\text{head}_1, \ldots, \text{head}_n)\mathbf{W}^O \\ \text{head}_j &= \text{Attention}\Big(\mathbf{Q}_1\mathbf{W}_j^Q, \mathbf{K}_1^a\mathbf{W}_j^K, \mathbf{V}_1^a\mathbf{W}_j^V\Big) \\ \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) &= \text{softmax}\Big(\mathbf{Q}\mathbf{K}^T/\sqrt{d_k}\Big)\mathbf{V}\end{aligned} \tag{2}$$

where $\mathbf{W}^O$, $\mathbf{W}_j^Q$, $\mathbf{W}_j^K$, and $\mathbf{W}_j^V$ imply the learnable parameter matrixes, concat$(\cdot)$, Attention$(\cdot)$, and softmax$(\cdot)$ denote the concatenation, self-attention, and softmax functions, and $\sqrt{d_k}$ indicates a scaling factor with a d_k value equals the dimension of $\mathbf{K}$. Compared with the standard MSA, the proposed SAMSA has lower computational complexity. For the input data $\mathbf{P} \in \mathbb{R}^{(h/s)\times(w/s)\times c'}$, the time complexities of MSA and

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SAMSA are

$$\Omega(\text{MSA}) = 4\frac{hw}{s^2}c'^2 + 2\left(\frac{hw}{s^2}\right)^2 c'$$
$$\Omega(\text{SAMSA}) = 2\left(\frac{hw+s\sqrt{hw}}{s^2}\right)c'^2 + 2\left(\frac{hw}{s^2}\right)^{3/2} c'. \quad (3)$$

The idea of reducing the computational costs of the transformer by decreasing the feature maps' spatial sizes can also be found in pyramid vision transformer (PVT) [52] and PVT V2 [53]. Here, we summarize the similar and different points between our SAVT block and the other two models. The similar point of the three models is that they reduce the sizes of $\mathbf{K}$ and $\mathbf{V}$ to whittle down the computational costs of the original visual transformer. However, the spatial reduction schemes are different from each other. For clarity, we use $\mathbf{K}$ as an example to explain. Suppose the size of $\mathbf{K}$ is $(h/s) \times (w/s) \times c'$. PVT uses a reshape and a linear projection to reduce its size with a fixed reduction ratio R. After this process, the size of the new feature map $\mathbf{K}'_{\text{PVT}}$ is $(h/sR) \times (w/sR) \times c'$. PVT V2 leverages the average pooling to reduce the size of $\mathbf{K}$ to $P \times P \times c'$ forcibly, where $P < (h/s)$ and $P < (w/s)$. In other words, the reduced feature map $\mathbf{K}'_{\text{PVTV2}}$ has a size of $P \times P \times c'$. In contrast to the two schemes above, the reduction strategy in our SAVT block utilizes average pooling and 1×1 convolution operations to achieve a spatial reduction of $\mathbf{K}$ in an adaptive way. The size of the new feature map $\mathbf{K}'_{\text{SAP}}$ is $\lceil\sqrt{h/s}\rceil \times \lceil\sqrt{w/s}\rceil \times c'$. This way, we do not need to tune the hyper-parameters (R or P) to control the reduced spatial size. Besides the above point, another difference is that we add a 1×1 convolution after the dimension-reduced K'_{SAP}. This seemingly inconspicuous operation can help SAVT regularize the information within the dimension-reduced feature maps. The benefits of our adaptive spatial reduction scheme and additional 1×1 convolution will be studied in Section IV-E1.

2) Full-Range Fusion Module: The targets of FRFM are twofold. One is to fuse the information (e.g., complex land covers and temporal clues) corresponding to the input data pair mutually. Here, the temporal clues mean the progression of content changes. The other one is to point out the salient clues hidden in the input data. To achieve these goals, there are two submodules in FRFM [see Fig. 5(a)], including a cross fusion submodule and a convolutional block attention submodule (CBAM) [54].

The schematic illustration of the cross-fusion submodule can be found in Fig. 5(b). When input $\{\mathbf{X}, \mathbf{Y}\} \in \mathbb{R}^{h\times w\times c}$ to FRFM, the cross fusion submodule first fuses their messages and generates $\{\mathbf{X}^{\text{cf}}, \mathbf{Y}^{\text{cf}}\} \in \mathbb{R}^{h\times w\times c}$. For $\mathbf{X}$, this process can be formulated as

$$\mathbf{X}^{\text{cf}} = \mathbf{X} + \left[\text{softmax}\left(\frac{\mathbf{Q}_y\mathbf{K}_x^T}{\sqrt{d_{kx}}}\right)\mathbf{V}_x\right]\mathbf{W}_x^O \quad (4)$$

where $\mathbf{W}_x^O$ indicates a learnable parameter matrix, $\sqrt{d_{kx}}$ is a scaling factor and the value of d_{kx} is the dimension of $\mathbf{K}_x$, $\mathbf{K}_x$ and $\mathbf{V}_x$ denote the key and value of $\mathbf{X}$, and $\mathbf{Q}_y$ implies the query of $\mathbf{Y}$. Here, the definitions of $\{\mathbf{Q}_y, \mathbf{K}_x, \mathbf{V}_x\}$ are

$$\mathbf{Q}_y = \mathbf{Y}\mathbf{W}_y^Q, \quad \mathbf{K}_x = \mathbf{X}\mathbf{W}_x^K, \quad \mathbf{V}_x = \mathbf{X}\mathbf{W}_x^V \quad (5)$$

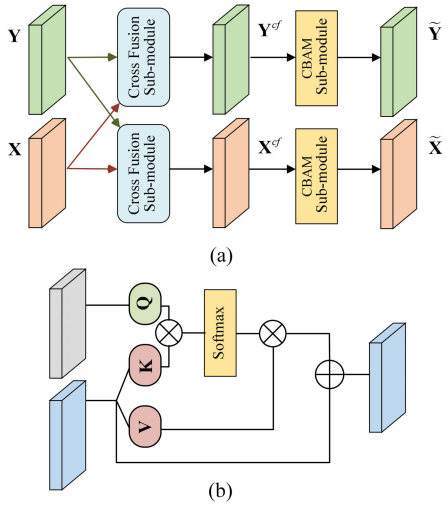


Fig. 5. Architecture of the FRFM and cross fusion submodule. (a) FRFM. (b) Cross fusion submodule.

where $\mathbf{W}_y^Q$, $\mathbf{W}_x^K$, and $\mathbf{W}_x^V$ are learnable weights. Similarly, the cross fusion process for $\mathbf{Y}$ is

$$\mathbf{Y}^{\text{cf}} = \mathbf{Y} + \left[\text{softmax}\left(\frac{\mathbf{Q}_x\mathbf{K}_y^T}{\sqrt{d_{ky}}}\right)\mathbf{V}_y\right]\mathbf{W}_y^O$$
$$\mathbf{Q}_x = \mathbf{X}\mathbf{W}_x^Q, \quad \mathbf{K}_y = \mathbf{Y}\mathbf{W}_y^K, \quad \mathbf{V}_y = \mathbf{Y}\mathbf{W}_y^V \quad (6)$$

where $\sqrt{d_{ky}}$ is a scaling factor and the value of d_{ky} is the dimension of $\mathbf{K}_y$, $\mathbf{K}_y$, and $\mathbf{V}_y$ denote the key and value of $\mathbf{Y}$, and $\mathbf{Q}_x$ implies the Query of $\mathbf{X}$.

Then, two CBAMs are applied to $\mathbf{X}^{\text{cf}}$ and $\mathbf{Y}^{\text{cf}}$ to explore the salient information from two separated aspects. CBAM links channel attention and spatial attention serially. For $\mathbf{X}^{\text{cf}}$, CBAM sequentially deduces a channel attention score $\mathbf{S}_x^c \in \mathbb{R}^{1\times 1\times c}$ and a spatial attention score $\mathbf{S}_x^s \in \mathbb{R}^{h\times w\times 1}$ with the assistance of average-pooling, max-pooling, MLP, and convolution. Next, the obtained scores are multiplied back to $\mathbf{X}^{\text{cf}}$ for producing the refined feature data $\tilde{\mathbf{X}}$. This procedure can be formulated as

$$\tilde{\mathbf{X}}^{\text{cf}} = \mathbf{S}_x^c\left(\mathbf{X}^{\text{cf}}\right) \otimes \mathbf{X}^{\text{cf}}$$
$$\tilde{\mathbf{X}} = \mathbf{S}_x^s\left(\tilde{\mathbf{X}}^{\text{cf}}\right) \otimes \tilde{\mathbf{X}}^{\text{cf}} \quad (7)$$

where $\otimes$ means the element-wise multiplication. Likewise, the CBAM process for $\mathbf{Y}^{\text{cf}}$ is

$$\tilde{\mathbf{Y}}^{\text{cf}} = \mathbf{S}_y^c\left(\mathbf{Y}^{\text{cf}}\right) \otimes \mathbf{Y}^{\text{cf}}$$
$$\tilde{\mathbf{Y}} = \mathbf{S}_y^s\left(\tilde{\mathbf{Y}}^{\text{cf}}\right) \otimes \tilde{\mathbf{Y}}^{\text{cf}} \quad (8)$$

where $\mathbf{S}_y^c \in \mathbb{R}^{1\times 1\times c}$ and $\mathbf{S}_y^s \in \mathbb{R}^{h\times w\times 1}$ denote the channel and attention scores. By exploring the salient messages, the changed/unchanged information within RS images can be highlighted. More details of CBAM can be referred to [54].

3) Structure of Siamese Encoder: Now, let us introduce the structure of our Siamese encoder. It comprises three stages serially, and each stage incorporates two SAVT blocks and an FRFM. When bi-temporal RS images $\{I_A, I_B\}$ with the size of $H \times W \times 3$ are input the encoder, the two SAVT blocks in the first stage will map them into feature maps $\mathbf{F}_1^A \in \mathbb{R}^{(H/4)\times(W/4)\times C_1}$ and $\mathbf{F}_1^B \in \mathbb{R}^{(H/4)\times(W/4)\times C_1}$. Then, $\{\mathbf{F}_1^A, \mathbf{F}_1^B\}$

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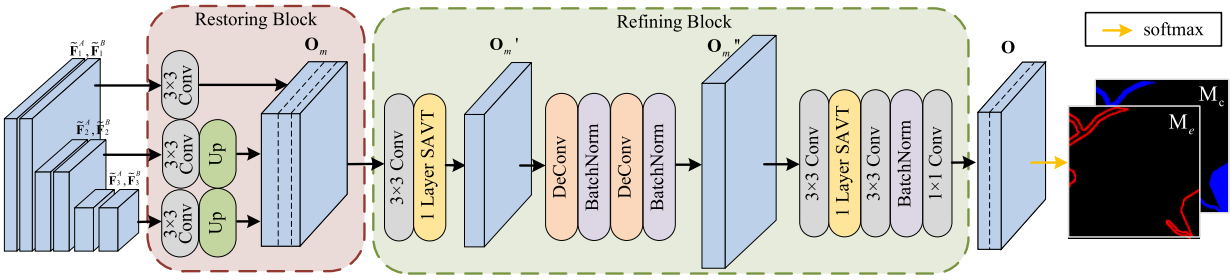


Fig. 6. Architecture of the restoring block and the refining block.

will be fed into the FRFM to generate the enhanced features $\{\tilde{\mathbf{F}}_1^A, \tilde{\mathbf{F}}_1^B\} \in \mathbb{R}^{(H/4)\times(W/4)\times C_1}$. By analogy, after the second and third stages, we can get $\{\mathbf{F}_2^A, \mathbf{F}_2^B, \tilde{\mathbf{F}}_2^A, \tilde{\mathbf{F}}_2^B\} \in \mathbb{R}^{(H/8)\times(W/8)\times C_2}$ and $\{\mathbf{F}_3^A, \mathbf{F}_3^B, \tilde{\mathbf{F}}_3^A, \tilde{\mathbf{F}}_3^B\} \in \mathbb{R}^{(H/16)\times(W/16)\times C_3}$ orderly. These feature maps involve not only the RS images' complex contents with various scales but also the temporal messages. Here, we set the number of channels C_1, C_2, and C_3 to be 64, 128, and 256, respectively. In addition, the stride s in the overlapping embedding scheme is set to be 4 for the first stage and is set to be 2 for the last two stages.

C. Edge-Aware Decoder

So far, we have obtained the encoded feature maps $\{\tilde{\mathbf{F}}_1^A, \tilde{\mathbf{F}}_2^A, \tilde{\mathbf{F}}_3^A, \tilde{\mathbf{F}}_1^B, \tilde{\mathbf{F}}_2^B, \tilde{\mathbf{F}}_3^B\}$ for bi-temporal RS images I_A and I_B. Then, they will be fed into the edge-aware decoder to produce the change map. Meanwhile, our decoder will generate an edge map to ensure the estimated changed areas have smooth and accurate edges. Along with the proposed joint loss function, the exact CD results can be obtained. The detailed framework of the edge-aware decoder is shown in Fig. 6, which incorporates a restoring block and a refining block.

1) Restoring Block: The goal of the restoring block is to integrate encoded feature maps' contributions. First, we concatenate the feature maps with the same scales in the channel dimension, i.e.,

$$\begin{aligned}\mathbf{O}_1 &= \left[\tilde{\mathbf{F}}_1^A; \tilde{\mathbf{F}}_1^B\right] \in \mathbb{R}^{\frac{H}{4}\times\frac{W}{4}\times 2C_1}\\ \mathbf{O}_2 &= \left[\tilde{\mathbf{F}}_2^A; \tilde{\mathbf{F}}_2^B\right] \in \mathbb{R}^{\frac{H}{8}\times\frac{W}{8}\times 2C_2}\\ \mathbf{O}_3 &= \left[\tilde{\mathbf{F}}_3^A; \tilde{\mathbf{F}}_3^B\right] \in \mathbb{R}^{\frac{H}{16}\times\frac{W}{16}\times 2C_3}.\end{aligned} \tag{9}$$

Then, three 3×3 convolutions are adopted to fuse the concatenated feature maps from the temporal aspects, i.e.,

$$\mathbf{O}_i' = \mathrm{Conv}_{3\times3}(\mathbf{O}_i), \quad i = 1, 2, 3 \tag{10}$$

where $\mathrm{Conv}_{3\times3}(\cdot)$ denotes the 3×3 convolution. Here, the spatial sizes of $\mathbf{O}_i'$ are the same as the $\mathbf{O}_i$, and the channel numbers $\mathbf{O}_i'$ are half of $\mathbf{O}_i$. After applying up-sampling operations to $\mathbf{O}_2'$ and $\mathbf{O}_3'$, the fused features $\mathbf{O}_m$ can be generated by concatenating them in the channel direction, and this process can be formulated as

$$\mathbf{O}_m = \left[\mathbf{O}_1', \mathrm{Up}\left(\mathbf{O}_2'\right), \mathrm{Up}\left(\mathbf{O}_3'\right)\right] \in \mathbb{R}^{\frac{H}{4}\times\frac{W}{4}\times(C_1+C_2+C_3)} \tag{11}$$

where $\mathrm{Up}(\cdot)$ means the up-sampling operations.

2) Refining Block: As the name suggests, this block is developed to improve the feature map $\mathbf{O}_m$ for CD. To further highlight multiscale information and temporal clues hidden in $\mathbf{O}_m$, we stack four 3×3 convolutions, one 1×1 convolution, two SAVTs (with only one SAVT encoder), two de-convolutions, and three batch-norms. The convolutions are used to capture the local knowledge, two SAVTs are adopted to analyze the global contextual clues, and de-convolutions are introduced to accomplish up-sampling with the consideration of relations between feature pixels. Particularly, $\mathbf{O}_m$ is first passed through a 3×3 convolution and a SAVT to generate $\mathbf{O}_m' \in \mathbb{R}^{(H/4)\times(W/4)\times(C_1+C_2+C_3)/2}$. Then, $\mathbf{O}_m'$ will be fed into two de-convolutions in sequence for producing $\mathbf{O}_m'' \in \mathbb{R}^{H\times W\times\lceil(C_1+C_2+C_3)/12\rceil}$. Note that two batch norms are followed by two de-convolutions, respectively. Finally, other two 3×3 convolutions, one 1×1 convolution, and a SAVT will be applied to $\mathbf{O}_m''$ for achieving the improved feature map $\mathbf{O} \in \mathbb{R}^{H\times W\times 2}$.

The reasons we leverage different tools to adjust the feature maps' sizes in two blocks are summarized as follows. First, the main goal of the restoring block is to fuse the encoded feature maps $\{\tilde{\mathbf{F}}_1^A, \tilde{\mathbf{F}}_1^B, \tilde{\mathbf{F}}_2^A, \tilde{\mathbf{F}}_2^B, \tilde{\mathbf{F}}_3^A, \tilde{\mathbf{F}}_3^B\}$. To this end, a simple and time-saving operation is needed to readily integrate all feature maps from the Siamese encoder. Therefore, we use up-sampling here to unify their spatial sizes. Second, the function of the refining block is to enhance the integrated feature map $\mathbf{O}_m$. To achieve this goal, the information within $\mathbf{O}_m$ should be analyzed carefully. Thus, de-convolution is adopted to adjust the feature map's size and explore the valuable clues within the feature map simultaneously.

3) Loss Function for Joint Learning: When we get $\mathbf{O}$, the final change map could be produced using a simple comparison scheme with a proper change loss function commonly. This way is feasible, however, it is not good at maintaining the edges of changed areas which would limit the accuracies of CD results. To overcome this limitation, we add an edge detection task to ensure that our model can be aware of the edges. To generate the change and edge predictions, we regard two channels of $\mathbf{O}$ as the confidence degrees of the change and edge information following the semantic segmentation ideas [55]. Thus, when $\mathbf{O}$ is mapped into $\mathbf{M}_s \in \mathbb{R}^{H\times W\times 2}$ by a softmax function, it can be split in the channel wise to generate a change map $\mathbf{M}_c \in \mathbb{R}^{H\times W\times 1}$ and an edge map $\mathbf{M}_e \in \mathbb{R}^{H\times W\times 1}$. Then, a joint loss function is proposed to narrow the gaps between them and the different ground truths. Considering the binarity of our tasks, the binary cross-entropy

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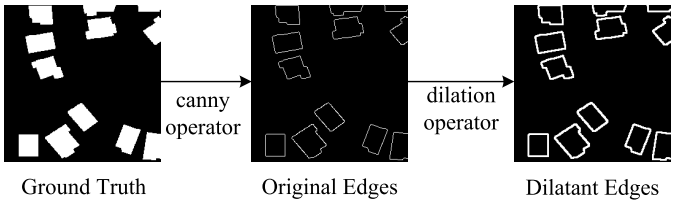


Fig. 7. Flowchart of generating edge ground truths.

(BCE) loss [56] is selected as the basic unit of our joint loss function. For the change map $\mathbf{M}_c$, the change loss is defined as

$$\mathcal{L}_c = -\mathbf{G}_c \log(\mathbf{M}_c) - (1 - \mathbf{G}_c) \log(1 - \mathbf{M}_c) \tag{12}$$

where $\mathbf{G}_c$ indicates the CD ground truth. For the edge map $\mathbf{M}_e$, the edge loss is specified as

$$\mathcal{L}_e = -\mathbf{G}_e \log(\mathbf{M}_e) - (1 - \mathbf{G}_e) \log(1 - \mathbf{M}_e) \tag{13}$$

where $\mathbf{G}_e$ denotes the edge detection ground truth. Finally, the joint loss function can be formulated as

$$\mathcal{L} = \lambda \mathcal{L}_c + (1 - \lambda)\mathcal{L}_e \tag{14}$$

where λ is a free parameter that controls the contributions of different terms.

There is another point we want to touch on, that is, the way for collecting the edge ground truth. Instead of annotating the edges of the changed areas manually, we can use $\mathbf{G}_c$ to generate $\mathbf{G}_e$ automatically. The schematic illustration of generating the edge ground truth can be found in Fig. 7. In detail, the Canny operator [57] is first applied to $\mathbf{G}_c$ for producing the original edges of the changed areas $\mathbf{G}_e^p$. Then, by passing $\mathbf{G}_e^p$ through a 3×3 convolution, the dilatant edges $\mathbf{G}_e^d$ can be obtained. In the following experiments, the dilatant and primary edge ground truths will be used to train our model sequentially. The main reason for collecting two sets of edge detection ground truths is to enable our EATDer to capture the edges of the changed areas in a coarse-to-fine way. The dilatant edge ground truth can help EATDer rapidly explore the edges of the changed areas. While the primary edge ground truth can assist EATDer in refining the obtained edges. Combining $\mathbf{G}_e^d$ and $\mathbf{G}_e^p$, the edges of changed areas in the final change maps are smooth and accurate. The impacts of $\mathbf{G}_e^d$ and $\mathbf{G}_e^p$ on our EATDer would be discussed in Section IV-E4.

IV. Experiment

A. Datasets Introduction

We select three CD datasets to evaluate the proposed EATDer, including the LEVIR building CD dataset (LEVIR-CD) [58], the CD dataset (CDD) proposed in [59], and the Wuhan University CD dataset (WHU) [60].

LEVIR-CD contains 637 Google Earth image patch pairs with a size of 1024×1024 pixels. The pixel resolution of those images is 0.5 m. LEVIR-CD mainly focuses on the changes in various buildings, such as villa residences and tall apartments. The original dataset is divided into 445 training pairs, 128 testing pairs, and 64 validation pairs. However, considering the limited computational capacity, we crop the patch pairs into 256×256 and get 7120 training pairs, 2048 testing pairs, and 1024 validation pairs according to the suggestions in [42]. CDD encompasses 16 000 season-varying RS image pairs collected from Google Earth. Their spatial sizes are 256×256, and their resolution varies from 0.03 to 1 m. The changed land covers are diverse in CDD, including buildings, cars, and roads. In agreement with the original literature, the numbers of images in the training, testing, and validation sets are 10 000, 3000, and 3000, respectively. WHU encloses a pair of RS images. Their spatial sizes and resolution are $32\,507 \times 15\,354$ and 0.2 m. The key objects in this dataset are buildings. Similar to [44], we first cut the images into 256×256 patches. Then, 5974, 744, and 743 image pairs are selected to construct the training, testing, and validation sets. The examples of three datasets and their change/edge detection ground truths are exhibited in Fig. 8.

B. Experimental Settings

The experiments are conducted using the Pytorch platform on a workstation with two GeForce RTX 3090 GPUs which have 24G Memory and 35.6 Tera floating point operations. Our EATDer is initialized with Xavier [61] to avoid the gradient vanishing and gradient explosion. The AdamW algorithm is adopted to optimize the parameters of our model. The learning rate, training epochs, and batch size are set to 0.001, 200, and 22. Note that we use the dilatant/original edge detection ground truths to train EATDer at the first and last 100 epochs. All RS images are resized into 256×256, and the common data augmentation strategies (e.g., flipping and rotation) are utilized to stabilize the model training. Three crucial free parameters should be decided in advance, i.e., the number of SAVT encoders L, the weight λ for two terms in (14), and the number of heads n in MSA [see (2)]. In the following experiments, we set $L = 4$, $\lambda = 0.3$, and $n = 4$ unless otherwise stated. Their influence on EATDer will be discussed in Section IV-F.

Here, we select precision (P), recall (R), $F1$-score ($F1$), and overall accuracy (OA) to achieve numerical assessments. Their definitions are

$$\begin{aligned} \mathrm{P} &= \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad \mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \\ F1 &= \frac{2 \times P \times R}{P + R}, \quad \mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN} + \mathrm{TN}} \end{aligned} \tag{15}$$

where TP, TN, FP, and FN imply the numbers of true positive, true negative, false positive, and false negative pixels, respectively.

C. Comparison With State-of-the-Art Methods

To study the performance of EATDer objectively, we compare it with the following nine methods, including SNUNet [18], ChangeNet [62], DARnet [21], DSAMNet [63], LightCDNet [64], ChangeFormer [46], BIT [42], ICIFNet [43], and CCLNet++ [65]. The first five methods are developed based on CNN, while the last four models are constructed based on the transformer. We complete these methods by using the released codes. Their hyper-parameters

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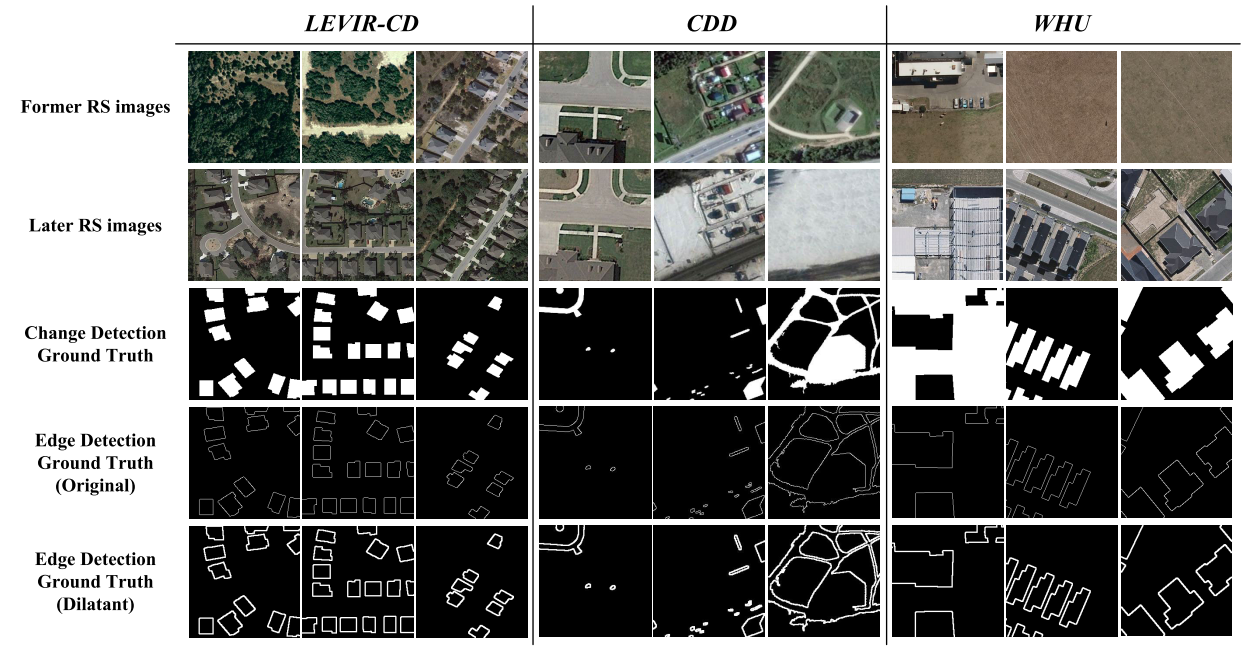


Fig. 8. Examples of three datasets. The former and later RS images are shown in the first two rows. The third row displays the CD ground truths. The original and dilatant edge detection ground truths are exhibited in the fourth and fifth rows.

TABLE I

PERFORMANCE OF EATDER AND COMPARED METHODS COUNTED ON THREE DATASETS. THE HIGHEST SCORES ARE MARKED IN **BOLD**

method	**LEVIR-CD** OA / R / P / F1 (%)	**CDD** OA / R / P / F1 (%)	**WHU** OA / R / P / F1 (%)	Params (M)	FLOPs (G)	Running Times (s)
SNUNet [18]	97.73 / 88.22 / 90.92 / 89.55	98.13 / 89.43 / 93.05 / 91.20	98.02 / 80.02 / 87.33 / 83.51	6.77	30.89	0.036
ChangeNet [62]	97.21 / 82.68 / 84.55 / 83.60	97.47 / 79.36 / 88.19 / 83.54	97.15 / 78.85 / 83.32 / 81.02	31.98	23.80	0.050
DARNet [21]	98.12 / 88.25 / 92.72 / 90.42	98.46 / 91.21 / 94.49 / 92.82	97.64 / 84.75 / 88.86 / 86.75	15.09	64.51	0.037
DSAMNet [63]	97.87 / 89.88 / 86.75 / 88.29	98.22 / 88.65 / 94.22 / 91.35	98.24 / 85.96 / 89.43 / 87.66	15.08	62.24	0.041
LightCDNet [64]	98.70 / 89.71 / 91.61 / 90.65	98.93 / 94.27 / 95.99 / 95.12	98.52 / 88.72 / 91.23 / 89.96	10.75	85.41	0.052
ChangeFormer [46]	98.17 / 87.38 / **93.81** / 90.48	98.16 / 90.87 / 94.95 / 92.86	98.08 / 85.63 / 90.89 / 88.18	16.98	45.97	0.081
BIT [42]	98.23 / 89.99 / 90.03 / 90.01	98.89 / 93.89 / 95.91 / 94.89	98.30 / 85.56 / **91.98** / 88.65	**3.55**	**4.35**	**0.027**
ICIFNet [43]	98.68 / 89.38 / 91.90 / 90.62	98.92 / 92.49 / 95.69 / 94.06	98.52 / 87.31 / 90.15 / 88.71	23.82	25.36	0.049
CCLNet++ [65]	98.74 / 90.00 / 91.72 / 90.85	98.95 / 93.92 / 96.24 / 95.07	98.54 / 88.68 / 89.91 / 89.29	28.78	46.55	0.043
EATDer (ours)	**98.75** / **90.67** / 91.74 / **91.20**	**98.97** / **95.13** / **96.83** / **95.97**	**98.58** / **88.74** / 91.32 / **90.01**	6.59	23.43	0.034

are set according to the original literature, and the experimental settings are the same as ours (see Section IV-B).

Table I summarizes the numerical results of different methods. From the observation of the table, we can find that the behavior of EATDer is stronger than that of the compared methods in most cases. Taking LEVIR-CD as an example, the improvements achieved by EATDer in $F1$ are 1.65% (over SNUNet), 7.60% (over ChangeNet), 0.78% (over DARnet), 2.91% (over DSAMNet), 0.55% (over LightCDNet), 0.72% (over ChangeFormer), 1.19% (over BIT), 0.58% (over ICIF-Net), and 0.35% (over CCLNet++). For another instance, the enhancements obtained by our model in OA counted on WHU are 0.56% (over SNUNet), 1.42% (over ChangeNet), 0.94% (over DARnet), 0.34% (over DSAMNet), 0.06% (over LightCDNet), 0.50% (over ChangeFormer), 0.28% (over BIT), 0.06% (over ICIF-Net), and 0.04% (over CCLNet++). These positive results owe to the following points. First, the stacked SAVT blocks in the Siamese encoder can explore the complex contents and their contextual relations from RS images, which helps EATDer understand RS images deeply. Second, the relationships of various land covers corresponding to RS images collected in different periods are considered by the proposed FRFM, which assists our model with the introduction of temporal clues. Third, the edge-aware decoder focuses on both change and edge detections, which ensures that EATDer can generate satisfactory change maps. Apart from the OA, P, R, and $F1$, we also count the parameters, FLOPS, and inference time costs of EATDer and other methods. It is easy to find that our EATDer is a relatively light and efficient model.

Despite the success mentioned above, we notice that P values of EATDer are not optimal in LEVIR-CD and WHU. This is mainly caused by the imbalanced sample problem. It is well-known that the number of changed samples is much less than that of the unchanged samples in normal CD datasets. Thus, the trained models are inclined to divide the pending pixels into the unchanged group. In this case, the obtained change maps would involve more FN pixels and fewer FP pixels, leading to larger P values and smaller R values (see ChangeFormer for LEVIR-CD and BIT for WHU). However, the distinct value gaps between P and R are improper, implying that the imbalanced sample issue impacts the models. Compared to them, although the P values of EATDer are slightly lower, the differences between P and R values are evidently

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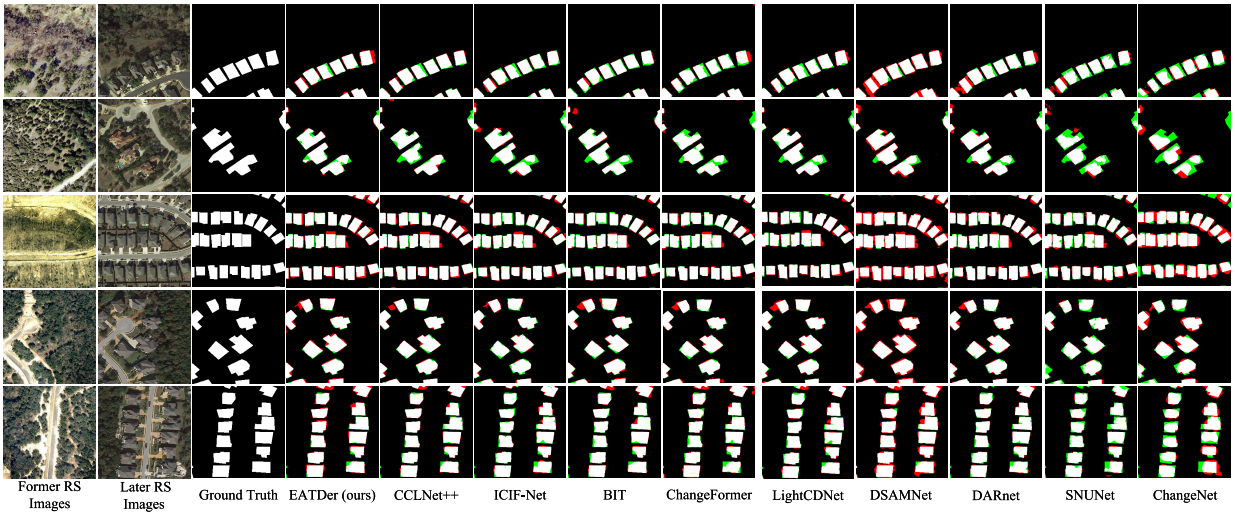


Fig. 9. Change maps obtained by different methods on the LEVIR-CD dataset. The bi-temporal RS images are displayed in the first two columns, and their CD ground truths are shown in the third column. The rest columns exhibit the produced change maps. For clarity, we mark TP, TN, FP, and FN pixels in white, black, red, and green, respectively.

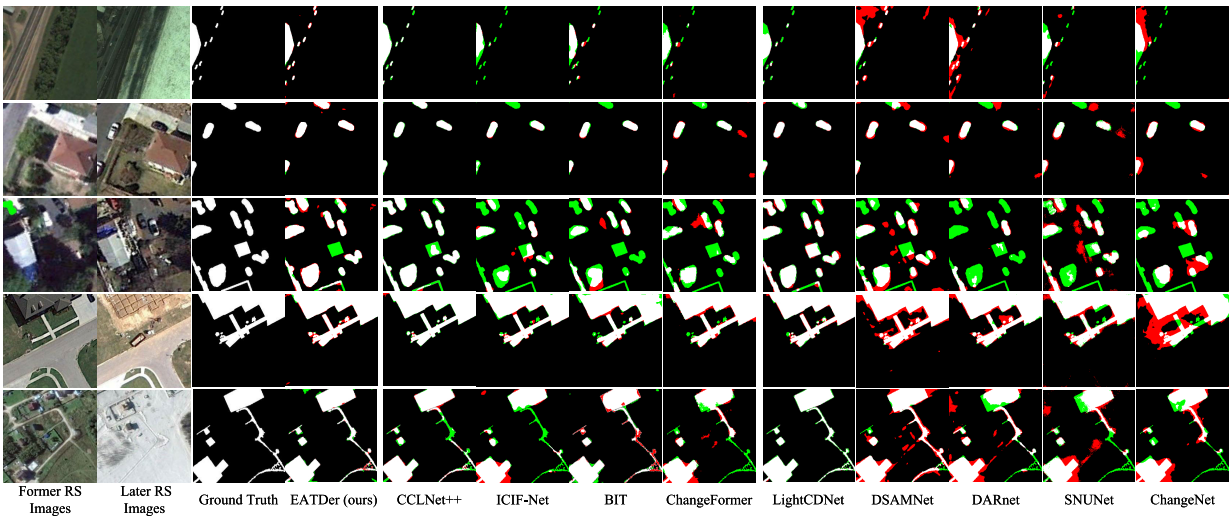


Fig. 10. Change maps obtained by different methods on the CDD dataset. The bi-temporal RS images are displayed in the first two columns, and their CD ground truths are shown in the third column. The rest columns exhibit the produced change maps. For clarity, we mark TP, TN, FP, and FN pixels in white, black, red, and green, respectively.

smaller, demonstrating that the proposed method can reduce the influence of sample imbalanced troubles.

To further study our EATDer, we randomly select five RS image pairs from each dataset, and their visual change maps obtained by different methods are shown in Figs. 9–11. For clarity, we use white, black, red, and green to represent TP, TN, FP, and FN pixels, respectively. By investigating them, we can find that the performance of the transformer-based models (ChangeFormer, BIT, ICIF-Net, and CCLNet++) is better than that of CNN-based approaches (SNUNet, ChangeNet, DARnet, DSAMNet, and LightCDNet). This is because the transformer-based models can capture both the local and global information hidden in RS images. Furthermore, the change maps generated by EATDer are the closest to the ground truths, which incorporate accurate change areas with clear edges. For example, in the third row of Fig. 9, the changed land covers (houses) in our change map are detected separately, and their edges are apparent. In the third row of Fig. 10, the numbers of FP and FN pixels in our change map are fewer than others. In the third row of Fig. 11, the changed land covers, which occupy large areas and are accessible to identify as background, can be detected by EATDer accurately. These encouraging observations confirm the effectiveness of the developed EATDer.

D. Rationality Analysis of Score Map Division

As introduced in Section III-C3, two tasks are developed and added at the top of the edge-aware decoder, i.e., change and edge detection. The CD task focuses on generating the change maps, while edge detection aims at refining the edges of the changed areas. To satisfy these two tasks, we first pass the decoder's output $\mathbf{O}_c \in \mathbb{R}^{H\times W\times 2}$ through a softmax function to get $\mathbf{O}_c \in \mathbb{R}^{H\times W\times 2}$, and then split it into a change map $\mathbf{M}_c \in \mathbb{R}^{H\times W\times 1}$ and an edge map $\mathbf{M}_e \in \mathbb{R}^{H\times W\times 1}$ from the channel dimension. To further study if this operation is useful or not, we remold our EATDer as follows. First, two 1×1 convolutions are applied to $\mathbf{O}$ for generating $\mathbf{O}_c \in \mathbb{R}^{H\times W\times 2}$ and $\mathbf{O}_e \in \mathbb{R}^{H\times W\times 2}$, which are used to reflect change

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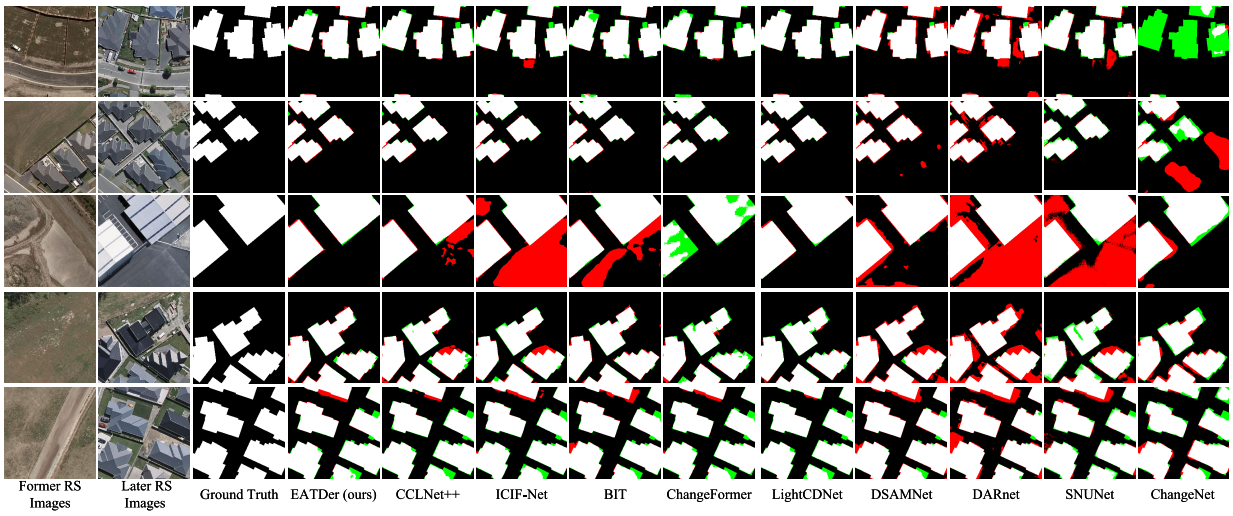


Fig. 11. Change maps obtained by different methods on the WHU dataset. The bi-temporal RS images are displayed in the first two columns, and their CD ground truths are shown in the third column. The rest columns exhibit the produced change maps. For clarity, we mark TP, TN, FP, and FN pixels in white, black, red, and green, respectively.

TABLE II

PERFORMANCE OF EATDER AND EATDER-V2 COUNTED ON THREE DATASETS (%)

Data Set	Model	OA	R	P	F1
LEVIR-CD	EATDer	98.75	90.67	91.74	91.20
	EATDer-v2	98.75	90.62	91.72	91.17
CDD	EATDer	98.97	95.13	96.83	95.97
	EATDer-v2	98.97	95.07	96.79	95.92
WHU	EATDer	98.58	88.74	91.32	90.01
	EATDer-v2	98.58	88.77	91.34	90.04

TABLE III

$F1$ SCORES (%), PARAMETER VOLUME, FLOPS, AND RUNNING TIMES OF EATDER AND ITS FOUR VARIETIES COUNTED ON THREE DATASETS

Methods	LEVIR-CD	CDD	WHU	Params (M)	FLOPs (G)	Running Times (s)
EATDer-T	90.42	95.96	89.91	9.54	29.66	0.288
EATDer-PVT	91.11	95.84	89.85	8.81	23.73	0.039
EATDer-PVTv2	91.10	95.94	89.92	6.61	23.48	0.037
EATDer-SwinT	90.72	94.89	89.13	6.63	23.74	0.041
EATDer	**91.20**	**95.97**	**90.01**	**6.59**	**23.43**	**0.034**

and edge information, respectively. Then, the change and edge maps $\mathbf{M}_c^{\text{new}} \in \mathbb{R}^{H\times W\times 2}$ and $\mathbf{M}_e^{\text{new}} \in \mathbb{R}^{H\times W\times 2}$ can be obtained by passing them through a softmax function. Next, the joint loss function closes the gaps between $\mathbf{M}_c^{\text{new}}/\mathbf{M}_e^{\text{new}}$ and the corresponding ground truths. We name this reconstructive model EATDer-v2. The performance of EATDer and EATDer-v2 counted on three datasets is summarized in Table II. By observing those results, we can find no distinct differences between the two detectors. This illustrates that directly dividing $\mathbf{O}$ into two elements channel-wise is feasible for our tasks.

E. Ablation Study

The behavior of our EATDer is decided by the following four essential factors: the SAVT block, FRFM, refining block, and edge clues. The first two factors belong to the Siamese encoder, while the last two are associated with the edge-aware decoder. To study them deeply, we conducted the following experiments. Note that we only adopt $F1$ as the assessment criterion in this section for clarity.

1) Ablation on SAVT Block: To verify the effectiveness of the proposed SAVT block, we replace it with the original transformer [66], PVT [52], PVT V2 [53], and SwinT [67] to construct EATDer-T, EATDer-PVT, EATDer-PVTv2, and EATDer-SwinT, respectively. For fairness, the number of heads in MSA mechanisms is set to four, and experimental settings are the same as the contents described in Section IV-B. The results of four new models are exhibited in Table III, and the results of the original EATDer are also displayed for reference. Besides the $F1$ scores, we count the Parameters, FLOPs, and inference time costs as well. We can find that our EATDer outperforms other models in all cases by investigating them. The reasons are 1) SAVT introduces the overlapping embedding scheme that benefits the exploration of local information and 2) SAVT remolds MSA by a simple SAP, which can reduce the time costs and parameter volume. The positive results discussed above confirm the usefulness of our SAVT block.

To further investigate the feature extraction capacity of SAVT, we randomly select three bi-temporal RS image pairs from LEVIR-CD and visualize the feature maps learned by EATDer in Fig. 12. Also, the feature maps obtained by EATDer-T are exhibited for reference. Observing these feature maps shows us that both the SAVT block and the transformer encoder can explore the significant clues hidden in temporal RS images for downstream CD. Taking the feature maps displayed in the second row as an example, the road in the former RS image and the buildings in the later RS images (which are the important messages for CD) can be captured by both EATDer and EATDer-T. This suggests that the proposed SAVT block possesses similar feature extraction capabilities to the transformer encoder. Considering the lower computational costs of the SAVT block, its superiorities are undeniable.

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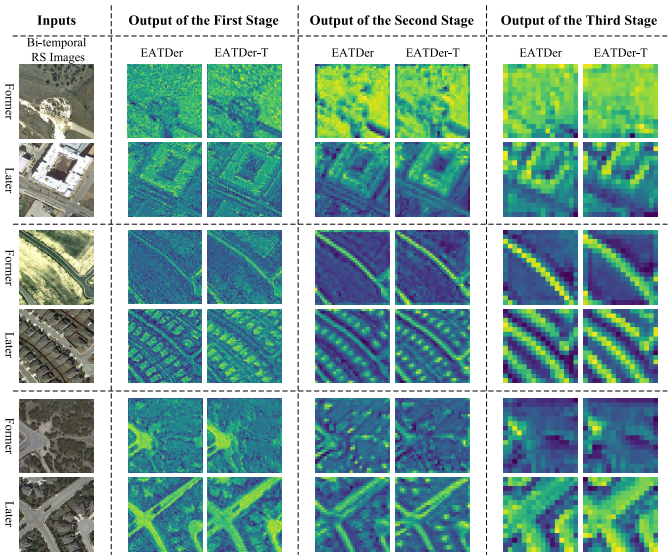


Fig. 12. Visualization of feature maps obtained by EATDer and EATDer-T. Three bi-temporal RS images are displayed in the first column, and the feature maps output in the different stages are exhibited in the last three columns orderly.

TABLE IV

$F1$-SCORE OF EATDER AND EATDER-PVTV2 COUNTED ON THREE DATASETS (%). THE SPATIAL SIZES INPUT RS IMAGES ARE VARIED FROM 512×512 TO 128×128

Spatial Sizes	Models	LEVIR-CD	CDD	WHU
512×512	EATDer-PVTv2	90.92	95.75	89.74
	EATDer	**91.11**	**95.87**	**89.92**
256×256	EATDer-PVTv2	91.10	95.94	89.92
	EATDer	**91.20**	**95.97**	**90.01**
128×128	EATDer-PVTv2	91.37	96.22	90.19
	EATDer	**91.38**	**96.23**	**90.21**

As discussed in Section III-B1, SAVT is similar to PVT V2 [53], which focuses on reducing the computational costs of the transformer by decreasing the feature maps' spatial sizes. However, there are two different points between SAVT and them. First, PVT V2 reduces the size of K and V to a fixed scale. Compared to it, the reduced size of K and V is adaptively decided by SAVT. The benefit of doing this is that it reduces the complexity of parameter tuning based on the size of input data and the requirements of downstream tasks and maintains good performance. Second, compared with PVT V2, SAVT adds a 1×1 convolution after the dimension-reduced K and V. This seemingly inconspicuous operation can help SAVT regularize the information within the dimension-reduced feature maps.

We do two groups of experiments to confirm the above statements. In the first group, we replaced all SAVT blocks with PVT V2 to construct EATDer-PVTv2. Then, RS images with different spatial sizes are input into EATDer and EATDer-PVTv2 to study the advantages of SAVT's adaptive dimension reduction scheme. Note that the free parameters of PVT V2 are consistent with the original literature. The sizes of RS images and $F1$ scores of RSCD results counted on three datasets are displayed in Table IV. By observing the results, we can find that EATDer outperforms EATDer-PVTv2 in all cases. This indicates that compared to a fixed dimension reduction mode, the adaptive dimension reduction strategy in SAVT can better adapt to the RSCD task. Admittedly, the reduced dimension (a free setting) of PVT V2 may be improper for our tasks, and we can find an empirically optimal dimension by careful tuning. However, the adaptive dimension reduction scheme makes SAVT more extensive compared with complicated tuning processes.

TABLE V

$F1$-SCORE OF EATDER, EATDER-W/O-1×1, EATDER-PVTV2, AND EATDER-PVTV2-W/-1×1 COUNTED ON THREE DATASETS (%)

Models	LEVIR-CD	CDD	WHU
EATDer	**91.20**	**95.97**	**90.01**
EATDer-w/o-1×1	91.16	95.96	89.99
EATDer-PVTv2	91.10	95.94	89.92
EATDer-PVTv2-w/-1×1	**91.12**	**95.94**	**89.95**

TABLE VI

$F1$ SCORES (%) OF EATDER AND ITS ABLATION VARIETIES COUNTED ON THREE DATASETS

Methods	LEVIR-CD	CDD	WHU
EATDer-no-FRFM	90.93	95.89	89.84
EATDer-1-FRFM	91.05	95.93	89.90
EATDer-2-FRFM	91.13	95.96	89.94
EATDer-no-Refine	90.12	94.65	89.04
EATDer-no-Edge	90.89	95.62	89.69
EATDer	**91.20**	**95.97**	**90.01**

In the second group, we remove (add) 1×1 convolution from (to) SAVT and PVT V2 to construct EATDer-w/o-1×1 and EATDer-PVTv2-w/-1×1. In this way, the function of the simple 1×1 convolution can be investigated. The $F1$ scores of different models counted on different datasets are exhibited in Table V. We can quickly discover that the behavior of EATDer and EATDer-PVTv2-w/-1×1 is stronger than that of EATDer-w/o-1×1 and EATDer-PVTv2. This implies the adoption of 1×1 convolution (after the dimension reduction) can normalize the knowledge of the dimension-reduced feature maps so that the final RSCD results can be improved.

2) Ablation on FRFM: As shown in Fig. 4, there are three stages in the Siamese encoder, and each stage has an FRFM to fuse the information from the bi-temporal RS images. Thus, we remove FRFMs individually to study their contributions to our EATDer deeply. In particular, we first remove the last FRFM and construct a new change detector named EATDer-2-FRFM. In this case, the outputs of two SAVT blocks $\{\mathbf{F}_3^A, \mathbf{F}_3^B\}$ and other enhanced feature maps $\{\tilde{\mathbf{F}}_1^A, \tilde{\mathbf{F}}_1^B, \tilde{\mathbf{F}}_2^A, \tilde{\mathbf{F}}_2^B\}$ will be fed into the edge-aware decoder to produce change maps. Analogously, we remove the last two FRFMs and construct EATDer-1-FRFM. In this specific scenario, the input for the edge-aware decoder is $\{\tilde{\mathbf{F}}_1^A, \tilde{\mathbf{F}}_1^B, \mathbf{F}_2^A, \mathbf{F}_2^B, \mathbf{F}_3^A, \mathbf{F}_3^B\}$. Finally, we remove all FRFM and establish EATDer-no-FRFM. In this instance, the outputs of SAVT blocks will be input into the decoder directly. The $F1$ scores of the above models counted on three datasets are exhibited in Table VI. Also, the results of EATDer are displayed for reference.

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By the observation of the results, we can find that the behavior of detectors becomes stronger with the number of FRFM increases. Taking the results counted on LEVIR-CD as an example, the $F1$ score of EATDer-no-FRFM is only 90.93%. When we add FRFMs gradually to detectors, the $F1$ score is increased to 91.13%. When three FRFMs are all added (i.e., EATDer), the $F1$ score is as high as 91.20%. This investigation means that FRFM plays a positive role in our EATDer. By embedding it into the Siamese encoder, not only can the valuable information corresponding to temporal RS images be fused, but also the salient clues hidden in RS images can be highlighted. Consequently, the CD results are improved.

3) Ablation on Refining Block: To study the significance of the refining block, we replace it with an up-sampling and a 3×3 convolution. The constructed model is named EATDer-no-Refine. Then, we use it to complete the CD on three datasets with the same experimental settings discussed in Section IV-B. Its $F1$ scores are also exhibited in Table VI. By comparing the results, we can find a distinct performance gap between EATDer-no-Refine and EATDer. Notably, compared with EATDer, the declines caused by removing the refining block are -1.08% (LEVIR-CD), -1.32% (CDD), and -0.97% (WHU). These discoveries demonstrate the importance of the refining block.

4) Ablation on Edge Clues: As mentioned in Section III-C3, we add an edge detection task to improve our CD results. To investigate if these clues are helpful or not, we fix the weight of $\mathcal{L}_e$ to zero and construct a new model recorded EATDer-no-Edge. According to the experimental settings mentioned in Section IV-B, we count the $F1$ scores of EATDer-no-Edge and EATDer on three datasets which can be found in Table VI. By observing them, we can find that EATDer outperforms EATDer-no-Edge in all cases, which implies the effectiveness of edge clues. Furthermore, we also display some change maps obtained by EATDer and EATDer-no-Edge in Fig. 13. Three bi-temporal RS image pairs are selected from LEVIR-CD (the first row), CDD (the second row), and WHU (the third row) randomly. From the observation of these maps, we can find that the maps generated by EATDer are closer to ground truths, and the changed areas' edges are continuous and apparent. These visual results confirm the strength of edge clues.

Apart from the above discussion, we continue to study the contributions of the primary edge ground truth $\mathbf{G}_e^p$ and dilatant edge ground truth $\mathbf{G}_e^d$, individually. To this end, we first only use $\mathbf{G}_e^p$ to complete the training. In this case, the trained EATDer is recorded as EATDer-PriEdge. Then, $\mathbf{G}_e^d$ and $\mathbf{G}_e^p$ are both used to train EATDer. As explained in Section III-C3, $\mathbf{G}_e^d$ is obtained by passing $\mathbf{G}_e^p$ through a 3×3 convolution. Here, we want to study if the 3×3 convolution is a proper choice for producing dilatant edge ground truth. Hence, we change the convolution kernel size from 3×3 to 9×9 with an interval of 2×2, and record different EATDers as EATDer3, EATDer5, EATDer7, and EATDer9, respectively. The $F1$ scores of different models counted on three datasets are depicted in Fig. 14. By observing the bars, we can find that combining $\mathbf{G}_e^d$ and $\mathbf{G}_e^p$ is beneficial to CD tasks in most cases, which implies the necessity of collecting the dilatant edges. However, another investigation is that when the size of convolutional kernels increases, the behavior of EATDer becomes weaker. This indicates that excessively inflated edges will generate unfavorable information for CD tasks, thereby reducing the accuracy of change maps. According to the results, the 3×3 convolution is a proper choice for producing dilatant edge ground truth.

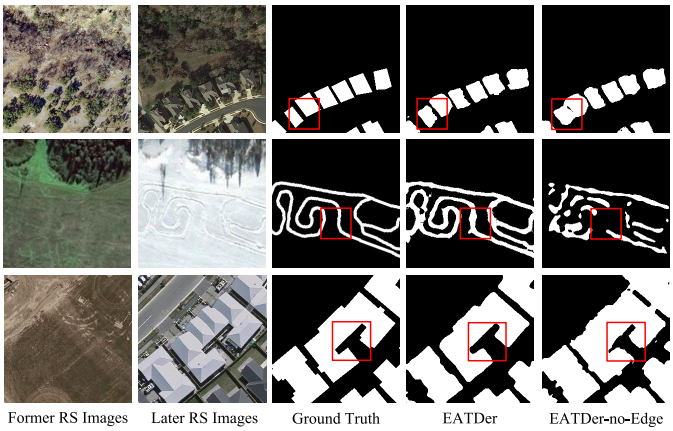


Fig. 13. Change maps of EATDer and EATDer-no-Edge. The bi-temporal RS images and their ground truths are exhibited in the first three columns. The change maps are displayed in the last two columns.

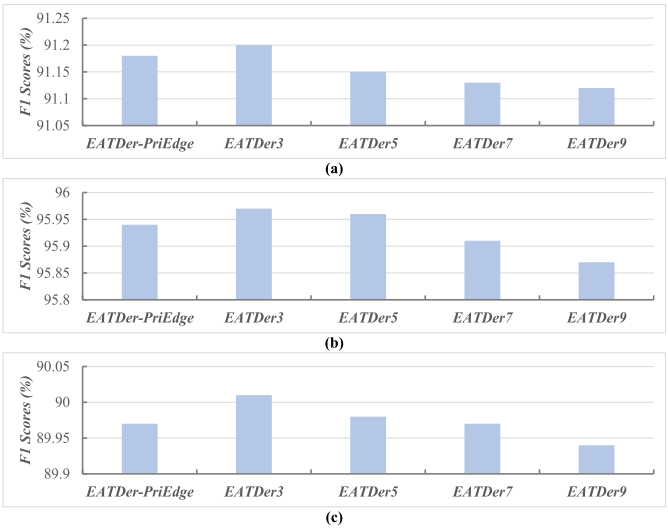


Fig. 14. $F1$ scores of EATDers with different edge ground truths. (a) LEVIR-CD. (b) CDD. (c) WHU.

F. Parameter Analysis

As discussed in Section IV-B, three free parameters should be decided in advance, including the number of SAVT encoders L, the weight λ in the loss function, and the number of heads in MSA n. Here, we study their influence on EATDer by altering their values.

For L, we fix $\lambda = 0.3$, $n = 4$, and change its values from 2 to 8. The $F1$ scores of EATDers with various numbers of SAVT encoders are displayed in Fig. 15. It is not surprising that the more SAVT encoders, the stronger EATDer. This is because the SAVT encoder focuses on exploring the local and global information within RS images. More SAVT encoders bring much richer knowledge that benefits downstream tasks.

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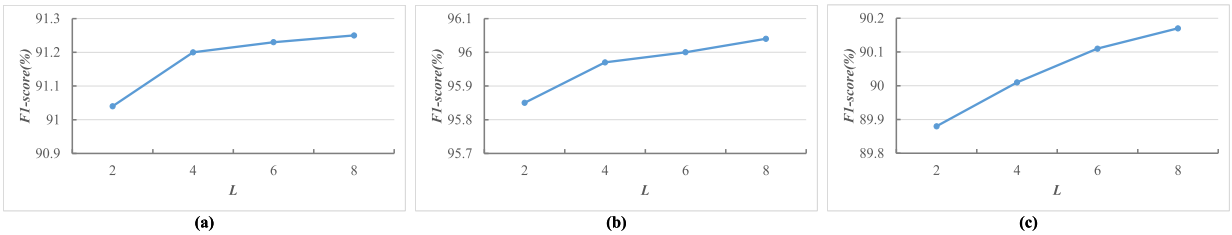


Fig. 15. $F1$ scores of EATDers with various numbers of SAVT encoders L. (a) LEVIR-CD. (b). CDD (c) WHU.

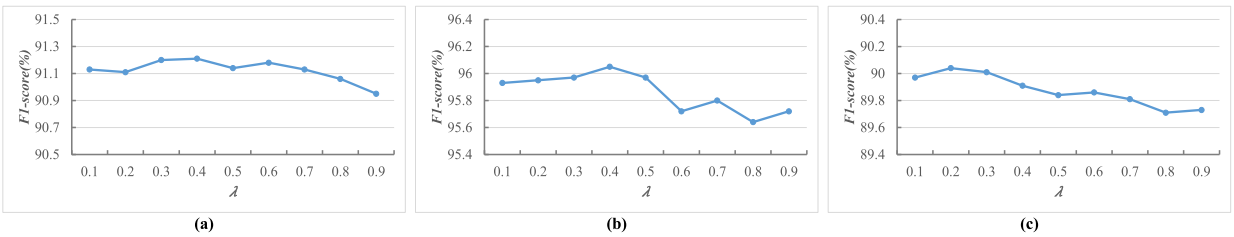


Fig. 16. $F1$ scores of EATDers with different λ. (a) LEVIR-CD. (b) CDD. (c) WHU.

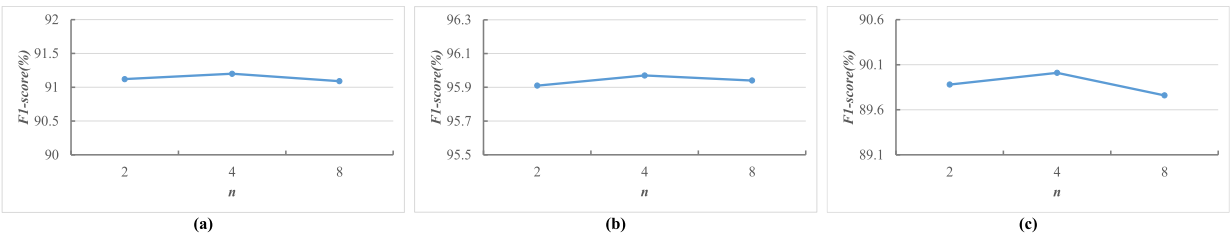


Fig. 17. $F1$ scores of EATDers with different n. (a) LEVIR-CD. (b) CDD. (c) WHU.

TABLE VII
PARAMETER VOLUME AND FLOPS OF EATDERS WITH VARIOUS NUMBERS OF SAVT ENCODERS

L	2	4	6	8
Params (M)	5.26	6.59	7.95	9.28
FLOPs (G)	20.73	23.43	26.19	28.92

However, the parameters of our model will also explode when more SAVT encoders are adopted. Therefore, we suggest $L = 4$ considering both performance and efficiency. Apart from the $F1$ scores, we also count the parameter volume and FLOPs of different EATDers, which are displayed in Table VII. Not surprisingly, the parameter volume and FLOPs of EATDer are both raised distinctly when L increases. Especially when $L > 4$, the increase in the parameter volume and FLOPs is much greater than the increase in the $F1$ score. Consequently, we suggest $L = 4$ considering both performance and efficiency.

For λ, we settle $L = 4$, $n = 4$, and vary λ from 0.1 to 0.9 with an interval of 0.1. The $F1$ scores of EATDers with different λ are summarized in Fig. 16. The trend of the three curves is ascending slightly before declining intensively, and their optimal ranges are $[0.3, 0.4]$ (LEVIR-CD), $[0.3, 0.5]$ (CDD), and $[0.2, 0.3]$ (WHU), respectively. These investigations imply the importance of the edge term $\mathcal{L}_e$ is little bigger than that of $\mathcal{L}_c$. To sum up, we suggest $\lambda = 0.3$. If the readers apply our EATDer on some new datasets, these two parameters $\{L, \lambda\}$ could be tuned beginning with $\{4, 0.3\}$.

For n, its values are varied in a range of $[2, 4, 8]$. Note that when n changes, L and λ remain 4 and 0.3. The $F1$ score curves of various EATDers with different n counted on three datasets are exhibited in Fig. 17. We can find that the behavior of EATDers becomes stronger when n is varied from 2 to 4. Nevertheless, once n increases to 8, the performance of EATDers is whittled down. This indicates that when the number of heads increases (within a certain range), the inherent global relationships in the input data can be more thoroughly explored. However, when n exceeds a certain threshold, the resulting global relationships become less reliable due to excessive data fragmentation, leading to a decrease in model performance. In summary, $n = 4$ is a suitable setting for our model in CD tasks.

V. CONCLUSION

This article presents a new CD method (EATDer) for RS images. EATDer contains a Siamese encoder and an edge-aware decoder, and they aim to capture essential knowledge from RS images and generate change maps, respectively. The basic unit of the Siamese encoder is the SAVT block, developed under the vision transformer's framework. SAVT can obtain local and global information from RS images with respectable time costs by introducing the overlapping embedding scheme and remolding the MSA mechanism. We also developed FRFM to further explore the temporal knowledge among RS images. By embedding FRFMs into the Siamese encoder, the land covers within RS images collected in different periods can be discovered mutually, which helps the encoder to highlight the changed and unchanged areas. The

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edge-ware decoder consists of a restoring block and a refining block, which focus on fusing the multiscale features and enhancing their discrimination. Taking the edges of changed areas into account, both change and edge detection results are output by the decoder. Along with the joint loss function, high-quality change maps can be produced. The encouraging results counted on three widespread datasets confirm that the proposed EATDer is applicable to RSCD.

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